# Syllabus: Digital Circuits & System Design

| Module No | Title of Module | No of Hours | Approximate Weightage |
|---|---|---|---|
| 1 | Logic Gates and Boolean Algebra | 4 | 8 |
| 2 | Combinational Circuits using basic gates as well as MSI devices | 7 | 15 |
| 3 | Elements of Sequential Logic Design | 7 | 15 |
| 4 | Sequential Logic Design | 7 | 15 |
| 5 | Logic Families and Programmable Logic Devices | 7 | 15 |
| 6 | Introduction to VHDL | 7 | 15 |
| | **Total** | **39** | |

| |
|---|
| **80 / 39 = 2.05** |
| **2.05 * 4 = 8.2** |
| **2.05 * 7 = 14.35** |
| |

# Contents of Module 1

| Unit No. | Contents | Hrs. |
| --- | --- | --- |
| | Logic Gates and Boolean Algebra | **4** |
| 1 | Digital logic gates, Realization using NAND, NOR gates, Boolean Algebra, De Morgan's Theorem, SOP and POS representation, K Map up to four variables. | |

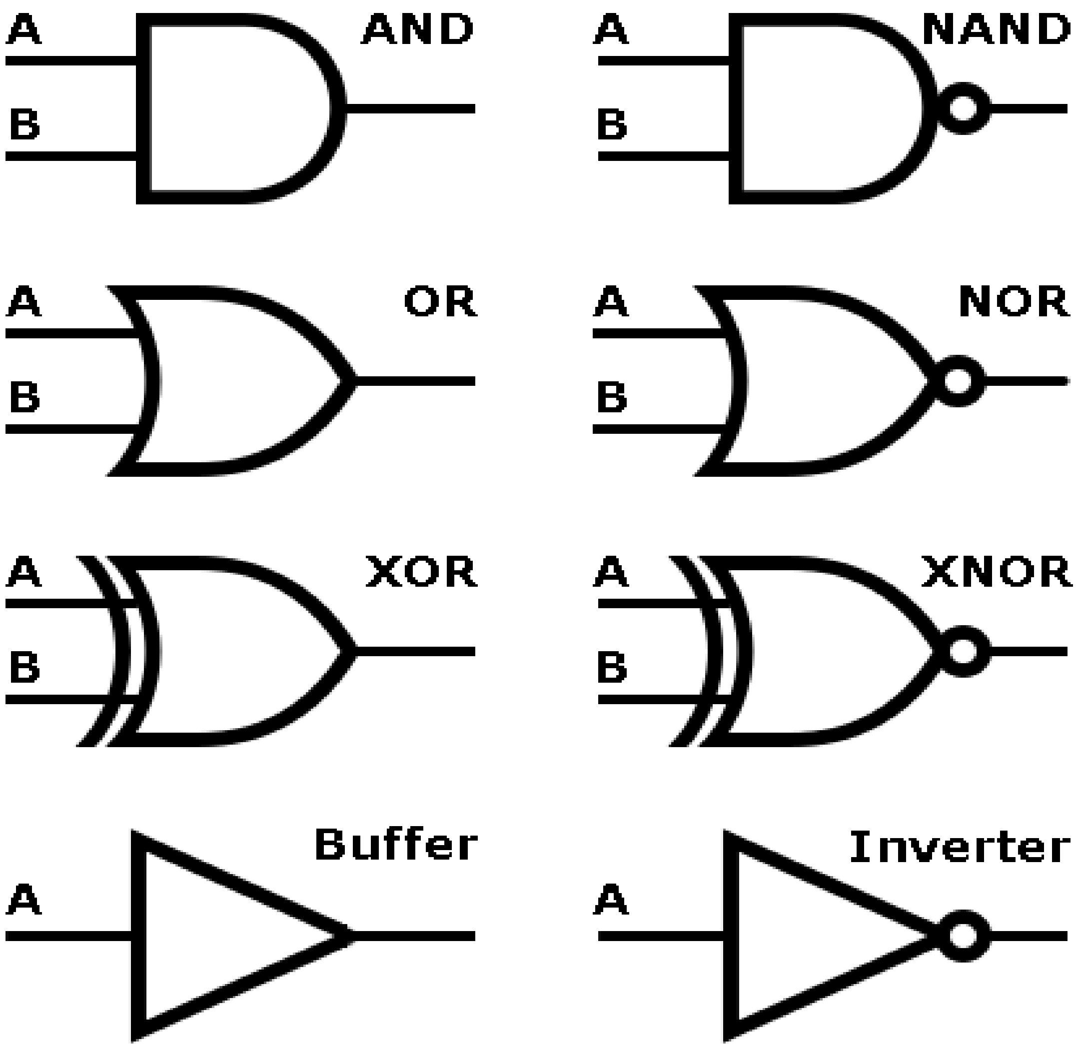
# Text Books:

| | |
|---|---|
| 1 | **R. P. Jain, Modern Digital Electronics, Tata McGraw Hill Education, Third Edition 2003.** |
| 2 | **Morris Mano, Digital Design, Pearson Education, Asia 2002.** |
| 3 | **J. Bhaskar, VHDL Primer, Third Edition, Prentice Hall, 1999.** |

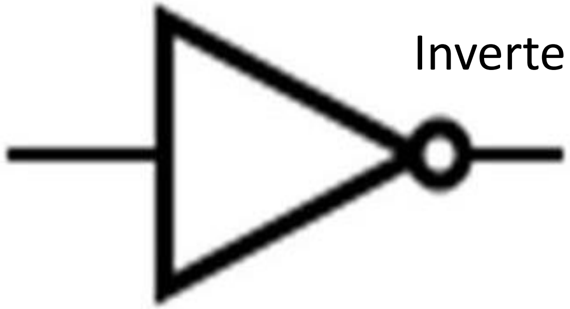
# Reference Books:

| | |
|---|---|
| 1 | Digital Logic Applications and Design – John M. Yarbrough, Thomson Publications, 2006 |
| 2 | John F. Warkerly, Digital Design Principles and Practices, Pearson Education, Fourth Edition, 2008. |
| 3 | Stephen Brown and ZvonkoVranesic, Fundamentals of digital logic design with Verilog design, McGraw Hill, 3rd Edition. |
| 4 | Volnei A. Pedroni,”Circuit Design with VHDL” MIT Press 2004. |
| 5 | Digital Circuits and Logic Design – Samuel C. Lee, PHI |
| 6 | William I. Flectcher, “An Engineering Approach to Digital Design”, Prentice Hall of India |
| 7 | Parag K Lala, “Digital System design using PLD”, BS Publications, 2003. |
| 8 | Charles H. Roth Jr., “Fundamentals of Logic design”, Thomson Learning, 2004 |

# Digital logic gates



# Digital logic gates

| GATE | SYMBOL | NOTATION | TRUTH TABLE |
|---|---|---|---|
| AND |  | $A \cdot B$ | |
| OR |  | $A + B$ | |
| NOT |  Inverter | $\overline{A}$ | |

| INPUT | | OUTPUT |
|---|---|---|
| A | B | A AND B |
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

| INPUT | | OUTPUT |
|---|---|---|
| A | B | A OR B |
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

| INPUT | OUTPUT |
|---|---|
| A | NOT A |
| 0 | 1 |
| 1 | 0 |

## Basic Logic Gates

There are **seven** basic logic gates: AND, OR, XOR, NOT, NAND, NOR, and XNOR

## Truth Table

Truth table gives all input possibilities and the corresponding outputs

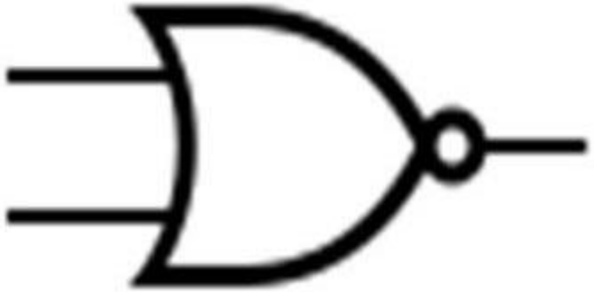
# Digital logic gates

1. Inverter or NOT gate:- Output of an inverter is complement of input. IC 7404 is TTL inverter IC.

2. AND gate :- Output of AND gate is high only if all inputs are high. If any input is low output is low. IC 7408 is TTL two input AND gate IC.

3. OR gate :- Output of OR gate is high if at least one inputs is high. If all inputs are  low output is low. IC 7432 is TTL two input OR gate IC.

# Digital logic gates

<table>
<tr>
<td>NAND</td>
<td></td>
<td>$\overline{A \cdot B}$</td>
<td>
<table>
<tr><th colspan="2">INPUT</th><th>OUTPUT</th></tr>
<tr><th>A</th><th>B</th><th>A NAND B</th></tr>
<tr><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td></tr>
</table>
</td>
</tr>
<tr>
<td>NOR</td>
<td></td>
<td>$\overline{A + B}$</td>
<td>
<table>
<tr><th colspan="2">INPUT</th><th>OUTPUT</th></tr>
<tr><th>A</th><th>B</th><th>A NOR B</th></tr>
<tr><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td></tr>
</table>
</td>
</tr>
<tr>
<td>XOR</td>
<td><br>EXOR</td>
<td>$A \oplus B$</td>
<td>
<table>
<tr><th colspan="2">INPUT</th><th>OUTPUT</th></tr>
<tr><th>A</th><th>B</th><th>A XOR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td></tr>
</table>
</td>
</tr>
</table>

# Digital logic gates

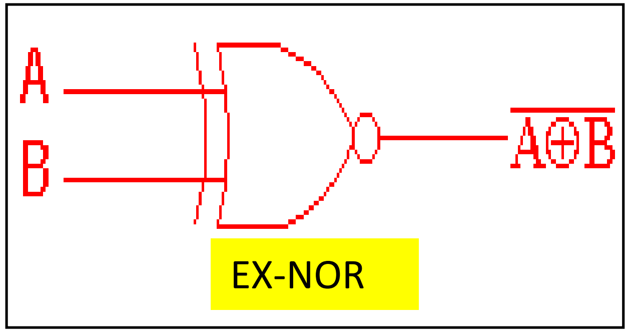
1. NAND gate :- Output of NAND gate is high if at least one inputs is low. If all inputs are  high output is low. IC 7400 is TTL two input NAND gate IC.

2. NOR gate :- Output of NOR gate is high if all inputs are low. If at least one input is high output is low. IC 7402 is TTL two input NOR gate IC.

3. EXOR gate :- Output of EXOR gate is high if odd number of inputs are high. If even number of inputs are high output is low. IC 7486 is TTL two input EXOR gate IC.

# EX-NOR Gate

The '**Exclusive-NOR'** gate circuit does the opposite to the EXOR gate. It will give a low output if **either, but not both**, of its two inputs are high. The symbol is an EXOR gate with a small circle on the output. The small circle represents inversion.




| 2 Input EXNOR gate | | |
|---|---|---|
| A | B | $\overline{A \oplus B}$ |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

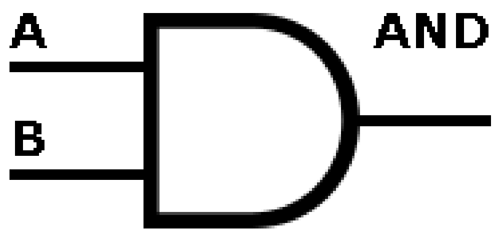
EXNOR gate :- Output of EXNOR gate is high if even number of inputs are high. If odd number of inputs are high output is low. IC 74266 is TTL two input EXNOR gate IC.

# Boolean Algebra

| Name | AND form | OR form |
|---|---|---|
| Identity law | $1A = A$ | $0 + A = A$ |
| Null law | $0A = 0$ | $1 + A = 1$ |
| Idempotent law | $AA = A$ | $A + A = A$ |
| Inverse law | $A\bar{A} = 0$ | $A + \bar{A} = 1$ |
| Commutative law | $AB = BA$ | $A + B = B + A$ |
| Associative law | $(AB)C = A(BC)$ | $(A + B) + C = A + (B + C)$ |
| Distributive law | $A + BC = (A + B)(A + C)$ | $A(B + C) = AB + AC$ |
| Absorption law | $A(A + B) = A$ | $A + AB = A$ |
| De Morgan's law | $\overline{AB} = \bar{A} + \bar{B}$ | $\overline{A + B} = \bar{A}\bar{B}$ |

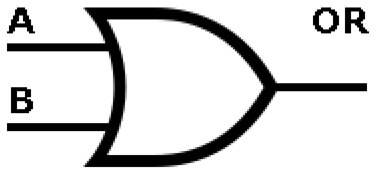
# Identity Law

1.A = A




| A | B | A.B |
|---|---|---|
| 0 | 1 | 0 |
| 1 | 1 | 1 |

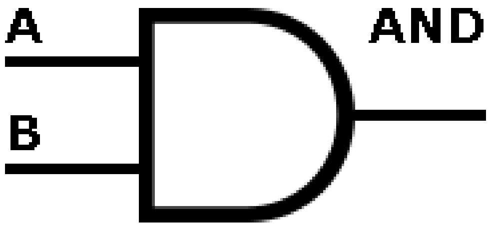
0 + A = A




| A | B | A + B |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |

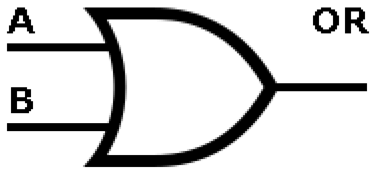
# Null Law

$0.A = 0$




| A | B | A.B |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 0 |

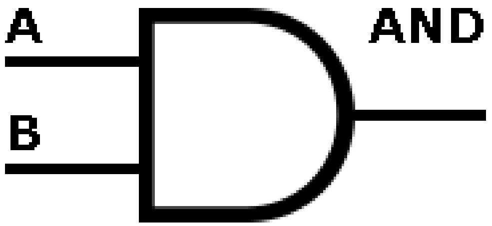
$1 + A = 1$




| A | B | A + B |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 1 | 1 |

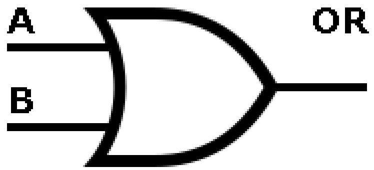
# Idempotent Law

$A.A = A$




| A | B | A.B |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 1 | 1 |

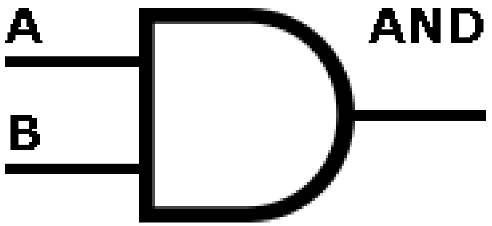
$A + A = A$




| A | B | A + B |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 1 | 1 |

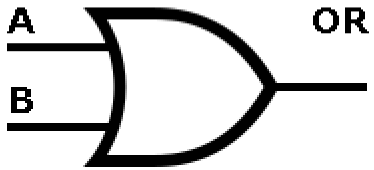
# Inverse Law

$A.\overline{A} = 0$




| A | B | A.B |
|---|---|---|
| 0 | 1 | 0 |
| 1 | 0 | 0 |

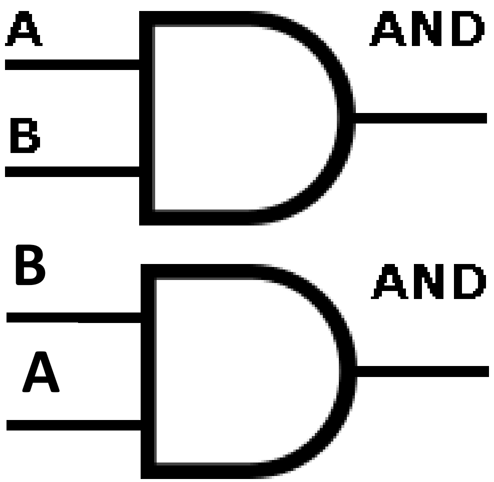
$A + \overline{A} = 1$



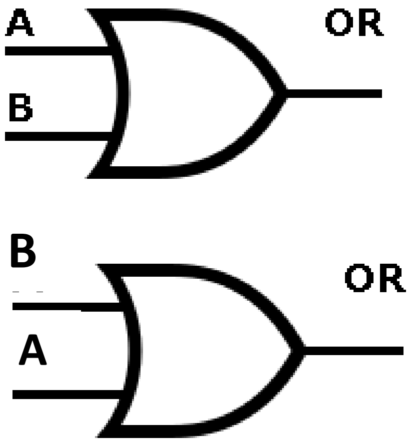

| A | B | A + B |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 0 | 1 |

# Commutative Law

$A.B = B.A$




$A + B = B + A$

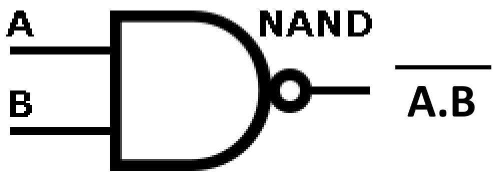


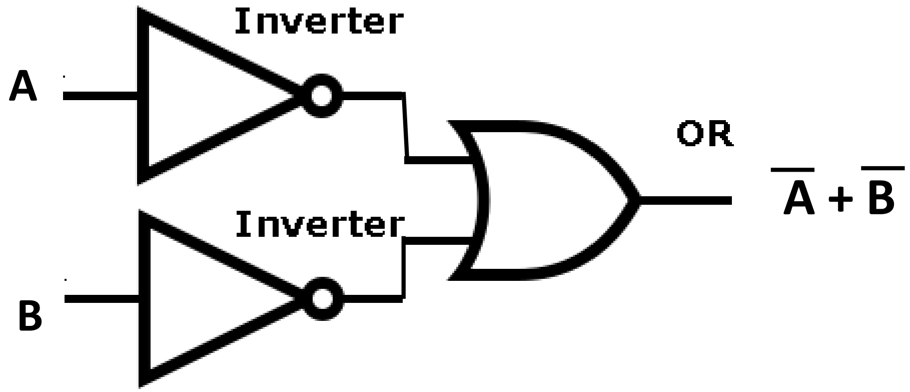

| A | B | O/P |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

| A | B | O/P |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

# De Morgan’s Theorem

$$\overline{A.B} = \overline{A} + \overline{B}$$



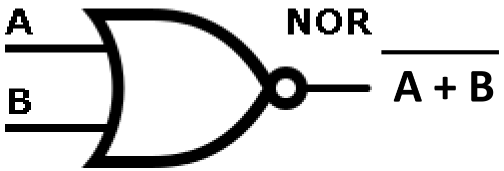

**TT to prove De Morgan’s first theorem**

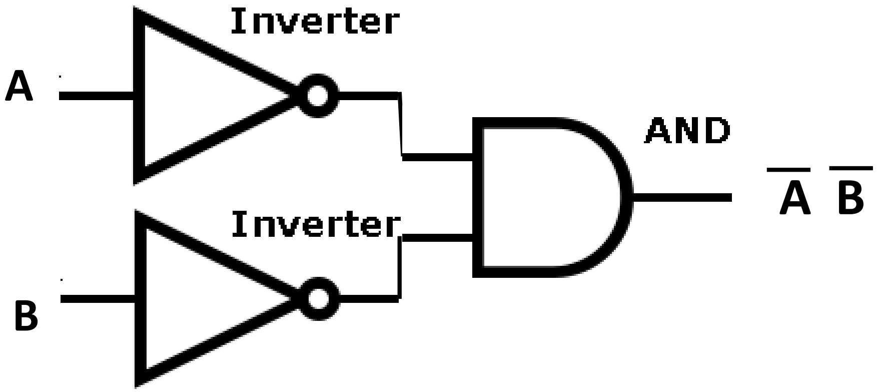
| A | B | $\overline{A}$ | $\overline{B}$ | $\overline{AB}$ | $\overline{A} + \overline{B}$ |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 | 0 |

**TT for $\overline{AB}$ and $\overline{A} + \overline{B}$ are same this proves De Morgan’s first theorem**

# De Morgan's Theorem

$$\overline{A + B} = \overline{A}\,\overline{B}$$




TT to prove De Morgan's second theorem

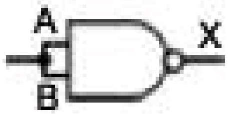
| A | B | $\overline{A}$ | $\overline{B}$ | $\overline{A + B}$ | $\overline{A}\ \overline{B}$ |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 |

TT for $\overline{A + B}$ and $\overline{A}\ \overline{B}$ are same this proves De Morgan's second theorem

# Universal Gates

## NAND, NOR Gates are called as Universal gates as other basic gates can be implemented using these gates

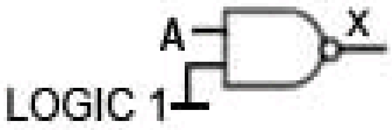
a.




NOT

| A | B | X |
| --- | --- | --- |
| 0 | 0 | 1 |
| 1 | 1 | 0 |

b.

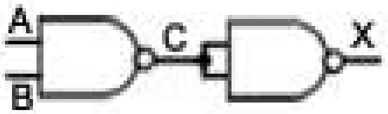



NOT

| A | B | X |
| --- | --- | --- |
| 0 | 1 | 1 |
| 0 | 1 | 1 |
| 1 | 1 | 0 |
| 1 | 1 | 0 |

Realization of inverter using NAND gate

c.




AND

| A | B | C | X |
| --- | --- | --- | --- |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

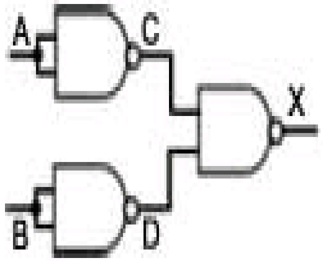
Realization of AND gate using NAND gates

# Universal Gates

## Realization of OR and NOR Gates using NAND gates

$$X = \overline{\overline{A}\,\overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$$

d.

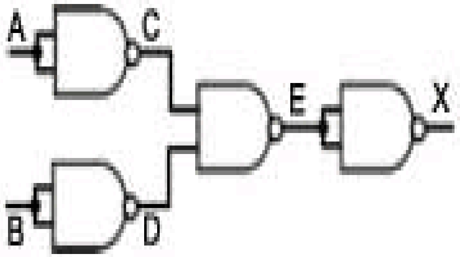



OR

| A | B | C | D | X |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |

Realization of OR gate using NAND gates

e.




NOR

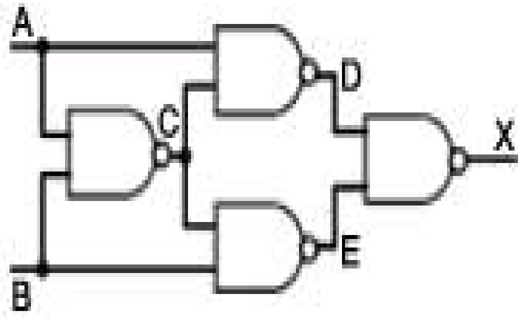
| A | B | C | D | E | X |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 |

Realization of NOR gate using NAND gates

# Universal Gates

## Realization of EXOR and EX-NOR Gates using NAND gates

f.

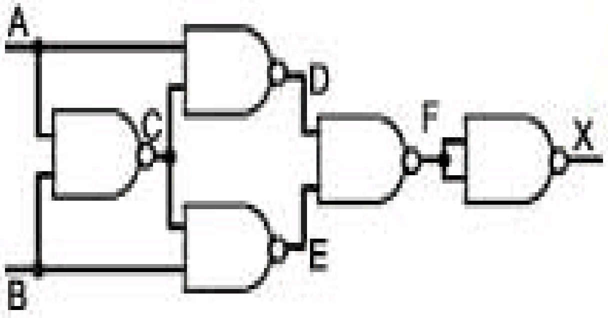



XOR

| A | B | C | D | E | X |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 1 | 1 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 | 0 |

Realization of EXOR gate using NAND gates

g.



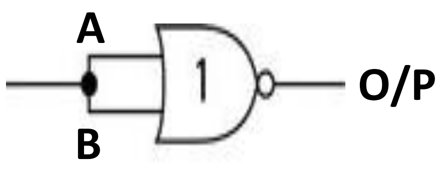

XNOR

| A | B | C | D | E | F | X |
|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 | 1 |

Realization of EX-NOR gate using NAND gates

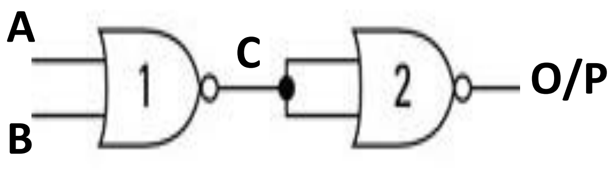
# Universal Gates

## Realization of Inverter and OR Gates using NOR gates




| A | B | O/P |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 0 |

Realization of Inverter using NOR gate

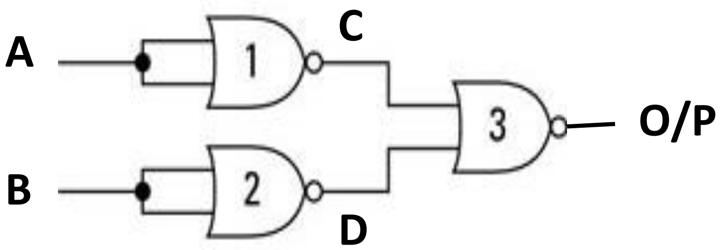



| A | B | C | O/P |
|---|---|---|---|
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 |

Realization of OR gate using NOR gates

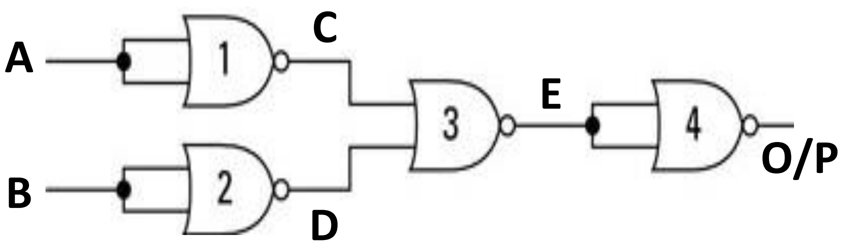
# Universal Gates

## Realization of AND and NAND Gates using NOR gates




| A | B | C | D | O/P |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 |

Realization of AND gate using NOR gates

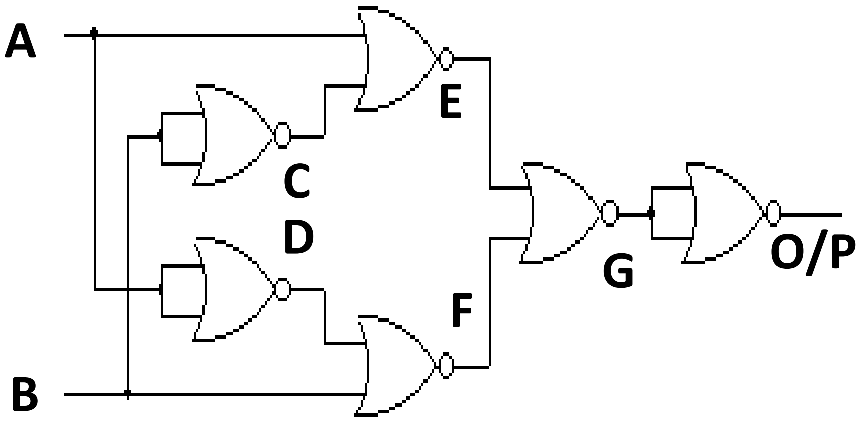



| A | B | C | D | E | O/P |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 | 0 |

Realization of NAND gate using NOR gates

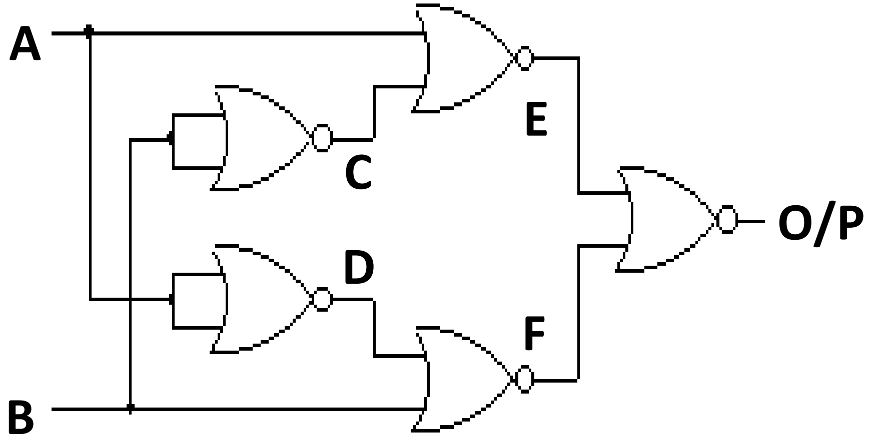
# Universal Gates

## Realization of EXOR and EX-NOR Gates using NOR gates




| A | B | C | D | E | F | G | O/P |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |

Realization of EXOR gate using NOR gates




| A | B | C | D | E | F | O/P |
|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 |

Realization of EX-NOR gate using NOR gates

# Karnaugh Maps

# Two Variable K-Map

## O/P Y is function of two variables A, B. (A MSB, B LSB)

Fundamental Products

| | $\overline{A}$ | $A$ |
|---|---|---|
| $\overline{B}$ | $\overline{A}\,\overline{B}$ | $A\,\overline{B}$ |
| $B$ | $\overline{A}\,B$ | $A\,B$ |

| | 0 | 1 |
|---|---|---|
| 0 | 00<br>0 | 10<br>2 |
| 1 | 01<br>1 | 11<br>3 |

Cell Numbers or Cell Positions

Complement of Variable means Zero.
True (Uncompleted) Variable means One.

# Three Variable K-Map

## O/P Y is function of three variables A, B and C. (A MSB, C LSB)

| | $\overline{A}$ | $A$ |
|---|---|---|
| $\overline{B}\,\overline{C}$ | $\overline{A}\,\overline{B}\,\overline{C}$ | $A\,\overline{B}\,\overline{C}$ |
| $\overline{B}\,C$ | $\overline{A}\,\overline{B}\,C$ | $A\,\overline{B}\,C$ |
| $B\,C$ | $\overline{A}\,B\,C$ | $A\,B\,C$ |
| $B\,\overline{C}$ | $\overline{A}\,B\,\overline{C}$ | $A\,B\,\overline{C}$ |

Fundamental Products

| | 0 | 1 |
|---|---|---|
| 00 | 0 | 4 |
| 01 | 1 | 5 |
| 11 | 3 | 7 |
| 10 | 2 | 6 |

Cell Numbers

# Four Variable K-Map

## O/P Y is function of Four variables A, B, C and D

| | $\bar{A}\bar{B}$ | $\bar{A}B$ | $AB$ | $A\bar{B}$ |
|---|---|---|---|---|
| $\bar{C}\bar{D}$ | | | | |
| $\bar{C}D$ | | | | |
| $CD$ | | | ABCD | |
| $C\bar{D}$ | | | | |

| | 0 0 | 0 1 | 1 1 | 1 0 |
|---|---|---|---|---|
| 0 0 | 0000 | 0100 | 1100 | 1000 |
| 0 1 | 0001 | 0101 | 1101 | 1001 |
| 1 1 | 0011 | 0111 | 1111 | 1011 |
| 1 0 | 0010 | 0110 | 1110 | 1010 |

# Four Variable K-Map

## O/P Y is function of Four variables A, B, C and D

| | 0 0 | 0 1 | 1 1 | 1 0 |
|---|---|---|---|---|
| 0 0 | **0** | **4** | **12** | **8** |
| 0 1 | **1** | **5** | **13** | **9** |
| 1 1 | **3** | **7** | **15** | **11** |
| 1 0 | **2** | **6** | **14** | **10** |

Cell Numbers

# Rules of K-Map

**1. Start from highest possible group**

In four variable map group of 16 will have highest priority. Second priority is for octets i.e. group of 8. Third priority is for quad (group of four). Fourth priority is for pair (group of two) and last priority is for single.

In three variable map group of 8 will have highest priority. Second priority is for quad. Third priority is for pair (group of two) and last priority is for single.

In two variable map group of 4 will have highest priority. Second priority is for pair and last priority is for single.

**2. In SOP (Sum of Products) consider groups of 1s only.**

**3. In SOP don’t care should be considered as 1 to have a bigger possible group.**

# Rules of K-Map

**4. The K-map can be rolled horizontally or vertically to have a bigger group**

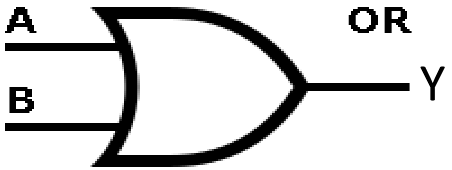
**5. Overlapping groups--- While encircling groups, use same 1 more than once to get larger group.**

**6. Redundant group--- Each group should have at least single one which is not part of any other group. A group whose all 1s are part of other groups is called as Redundant group. The redundant group should be avoided.**

**7. In POS (Product of Sum) consider 0s while taking groups. Consider don't care as zeroes to have a bigger group.**

## Truth Table of 2 input OR gate

| A | B | O/P Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |




$$Y = f\,(A, B) = \Sigma m\,(1, 2, 3)$$

| | $\overline{A}$ | $A$ |
|---|---|---|
| $\overline{B}$ | **0** | **1** |
| $B$ | 1 | 1 |

$A\,\overline{B} + A\,B$
$= A(\overline{B} + B)$
$= A$

$\overline{A}\,B + A\,B$
$= B(\overline{A} + A)$
$= B$

$$\therefore\ Y = A + B$$

# Truth Table of 2 input EXOR gate

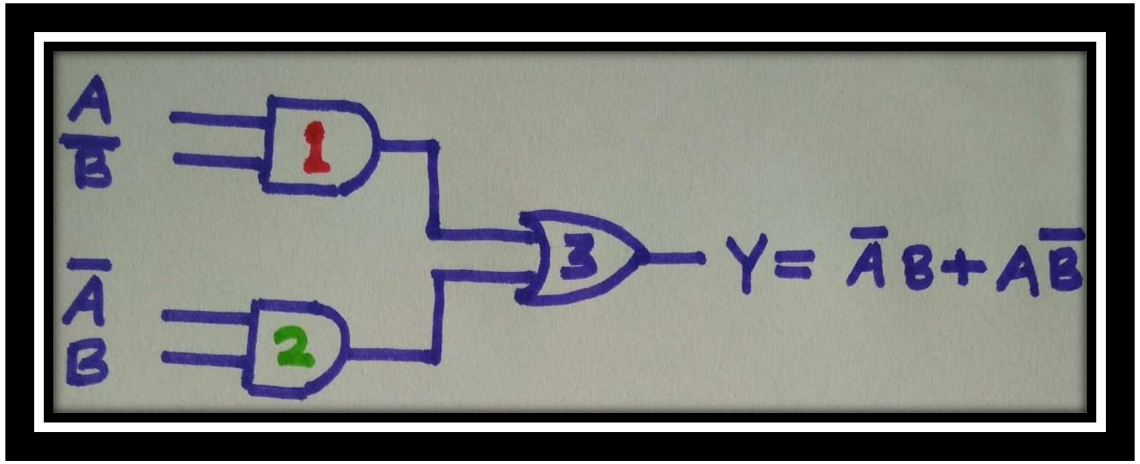
| A | B | O/P Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

$$Y = f\,(A, B) = \Sigma m\,(1, 2)$$

| | $\bar{A}$ | $A$ |
|---|---|---|
| $\bar{B}$ | **0** | **1** |
| $B$ | 1 | 0 |

$A\,\bar{B}$

$\bar{A}\,B$




$$\therefore\; Y = \bar{A}\,B + A\,\bar{B}$$

# Find the Simplest Boolean equation for the following function:

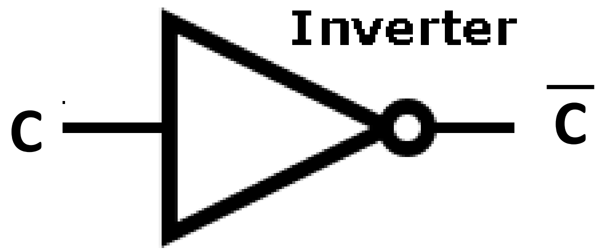
$$Y = f(A, B, C) = \Sigma m(0, 4, 6) + \Sigma d(1, 2, 5)$$

| | $\overline{A}$ | $A$ |
|---|---|---|
| $\overline{B}\,\overline{C}$ | 1 | 1 |
| $\overline{B}\,C$ | d | d |
| $B\,C$ | 0 | 0 |
| $B\,\overline{C}$ | d | 1 |

Rolling the map
$\overline{C}$

This is Redundant group

$\therefore\ Y = \overline{C}$



# Draw SOP Logic circuit for the following function:

$$Y = f(A, B, C) = \Sigma m(0, 6, 7) + \Sigma d(3, 4, 5)$$

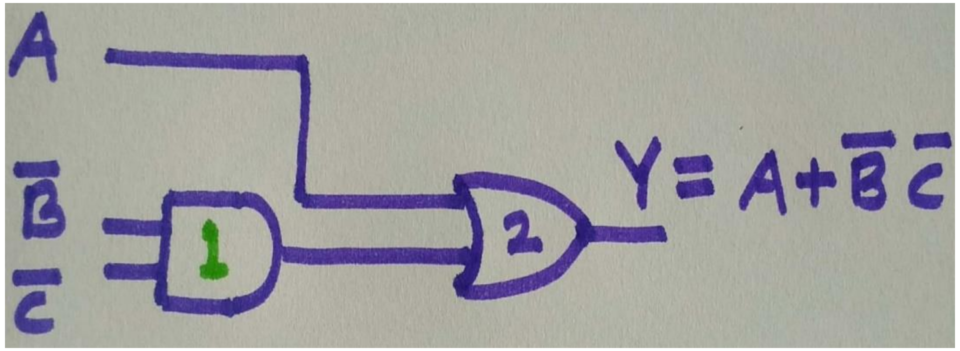
| | $\bar{A}$ | $A$ |
|---|---|---|
| $\bar{B}\,\bar{C}$ | 1 | d |
| $\bar{B}\,C$ | 0 | d |
| $B\,C$ | d | 1 |
| $B\,\bar{C}$ | 0 | 1 |

$\bar{B}\,\bar{C}$

$A$

This is Redundant group

$\therefore\ Y = A + \bar{B}\,\bar{C}$



# Draw SOP Logic circuit for the following function:

## $Y = f(A, B, C, D) = \Sigma m\,(7, 9, 10, 11, 12, 13, 14, 15)$

| | $\bar{A}\,\bar{B}$ | $\bar{A}\,B$ | $A\,B$ | $A\,\bar{B}$ |
|---|---|---|---|---|
| $\bar{C}\,\bar{D}$ | **0** | **0** | **1** | **0** |
| $\bar{C}\,D$ | **0** | **0** | **1** | **1** |
| $C\,D$ | **0** | **1** | **1** | **1** |
| $C\,\bar{D}$ | **0** | **0** | **1** | **1** |

AB

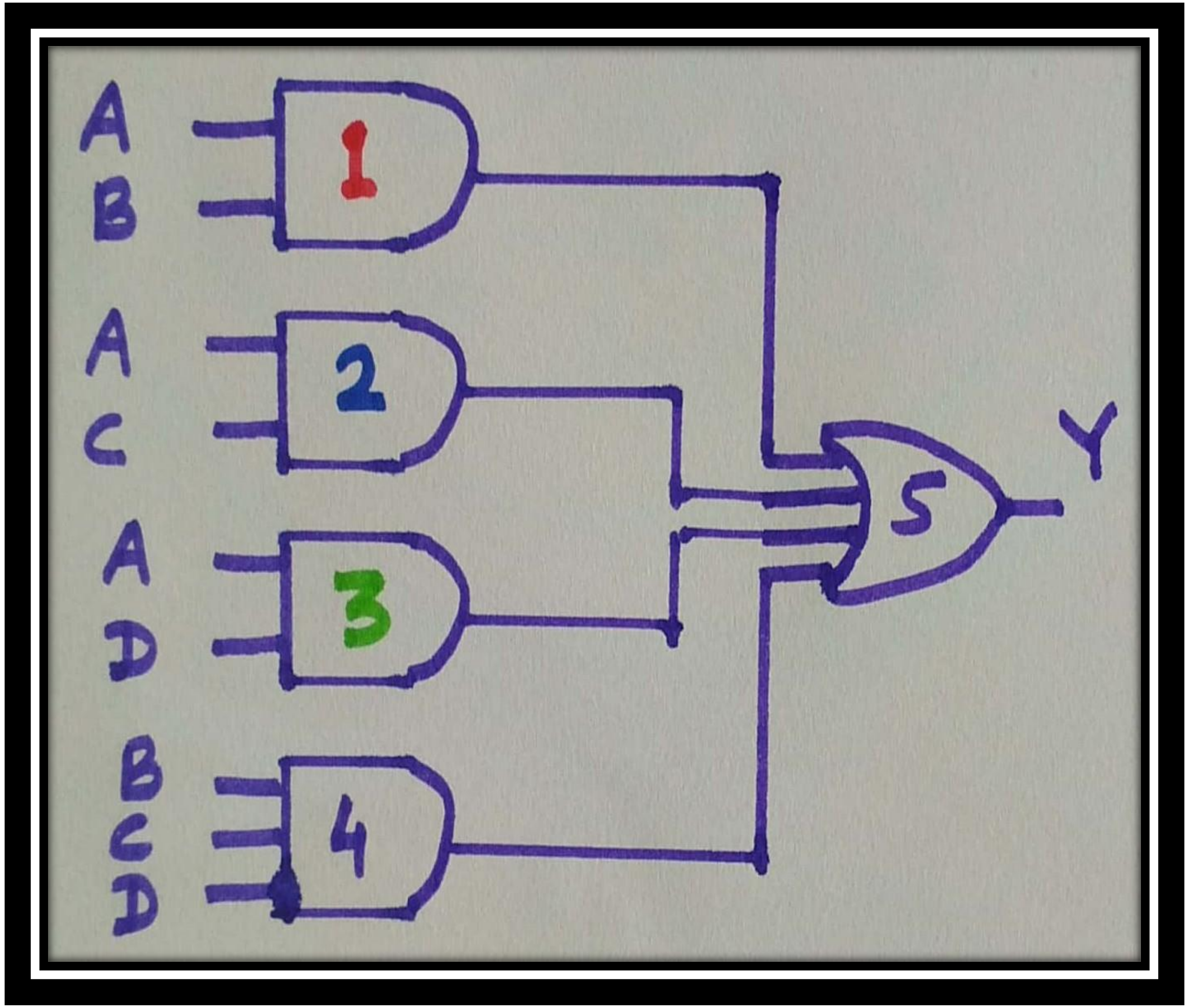
AD

AC

BCD

$\therefore\ Y = AB + AC + AD + BCD$

$\therefore\ Y = AB + AC + AD + BCD$



# Draw SOP Logic circuit for the following function:

$$Y = f(A, B, C, D) = \Sigma m\,(0, 3, 4, 5, 7) + \Sigma d\,(8, 9, 10, 11, 12, 13, 14, 15)$$

| | $\bar{A}\,\bar{B}$ | $\bar{A}\,B$ | $A\,B$ | $A\,\bar{B}$ |
|---|---|---|---|---|
| $\bar{C}\,\bar{D}$ | **1** | **1** | **d** | **d** |
| $\bar{C}\,D$ | **0** | **1** | **d** | **d** |
| $C\,D$ | **1** | **1** | **d** | **d** |
| $C\,\bar{D}$ | **0** | **0** | **d** | **d** |

$\bar{C}\,\bar{D}$

$B\,\bar{C}$

$B\,D$

$C\,D$

$B\bar{C}$ or $BD$ any one can be made redundant

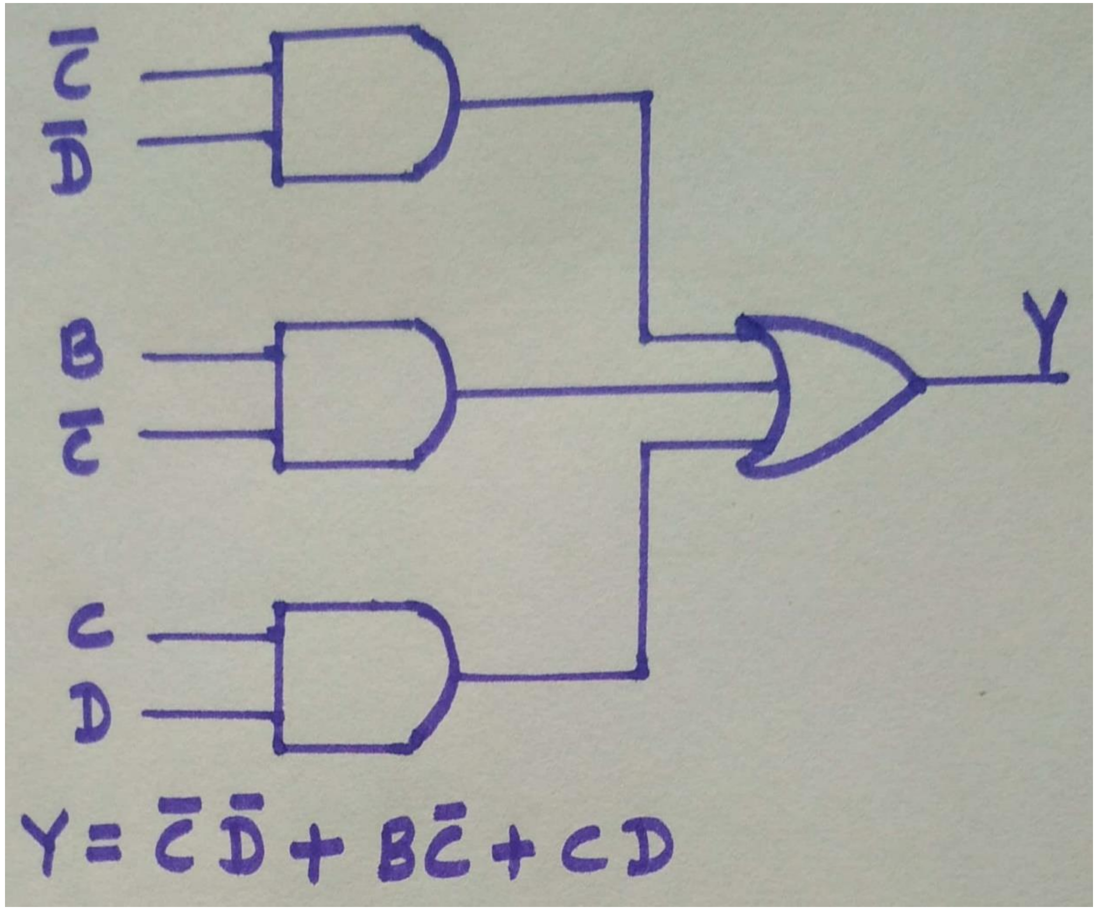
$$\therefore\ Y = \bar{C}\,\bar{D} + B\,\bar{C} + C\,D$$

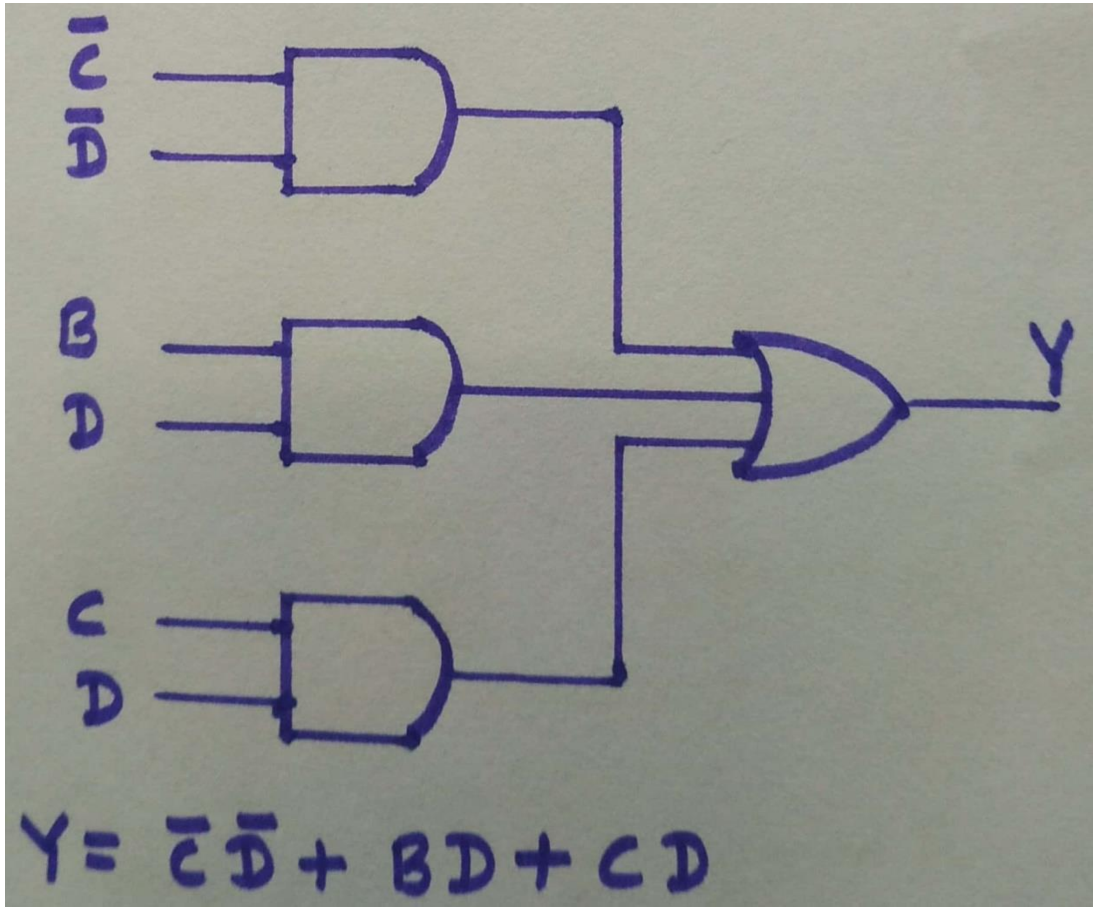
**OR**

$$\therefore\ Y = \bar{C}\,\bar{D} + B\,D + C\,D$$

# SOP Logic circuit for the function:

$$Y = f(A, B, C, D) = \Sigma m\,(0, 3, 4, 5, 7) + \Sigma d\,(8, 9, 10, 11, 12, 13, 14, 15)$$



# Draw SOP Logic circuit for the following function:

$$Y = f(A, B, C, D) = \Sigma m(0, 2, 3, 4, 5, 10, 12, 15) + \Sigma d(6, 8, 9)$$

| | $\bar{A}\,\bar{B}$ | $\bar{A}\,B$ | $A\,B$ | $A\,\bar{B}$ |
|---|---|---|---|---|
| $\bar{C}\,\bar{D}$ | **1** | **1** | **1** | **d** |
| $\bar{C}\,D$ | **0** | **1** | **0** | **d** |
| $C\,D$ | **1** | **0** | **1** | **0** |
| $C\,\bar{D}$ | **1** | **d** | **0** | **1** |

$\bar{C}\,\bar{D}$

$\bar{A}\,B\,\bar{C}$

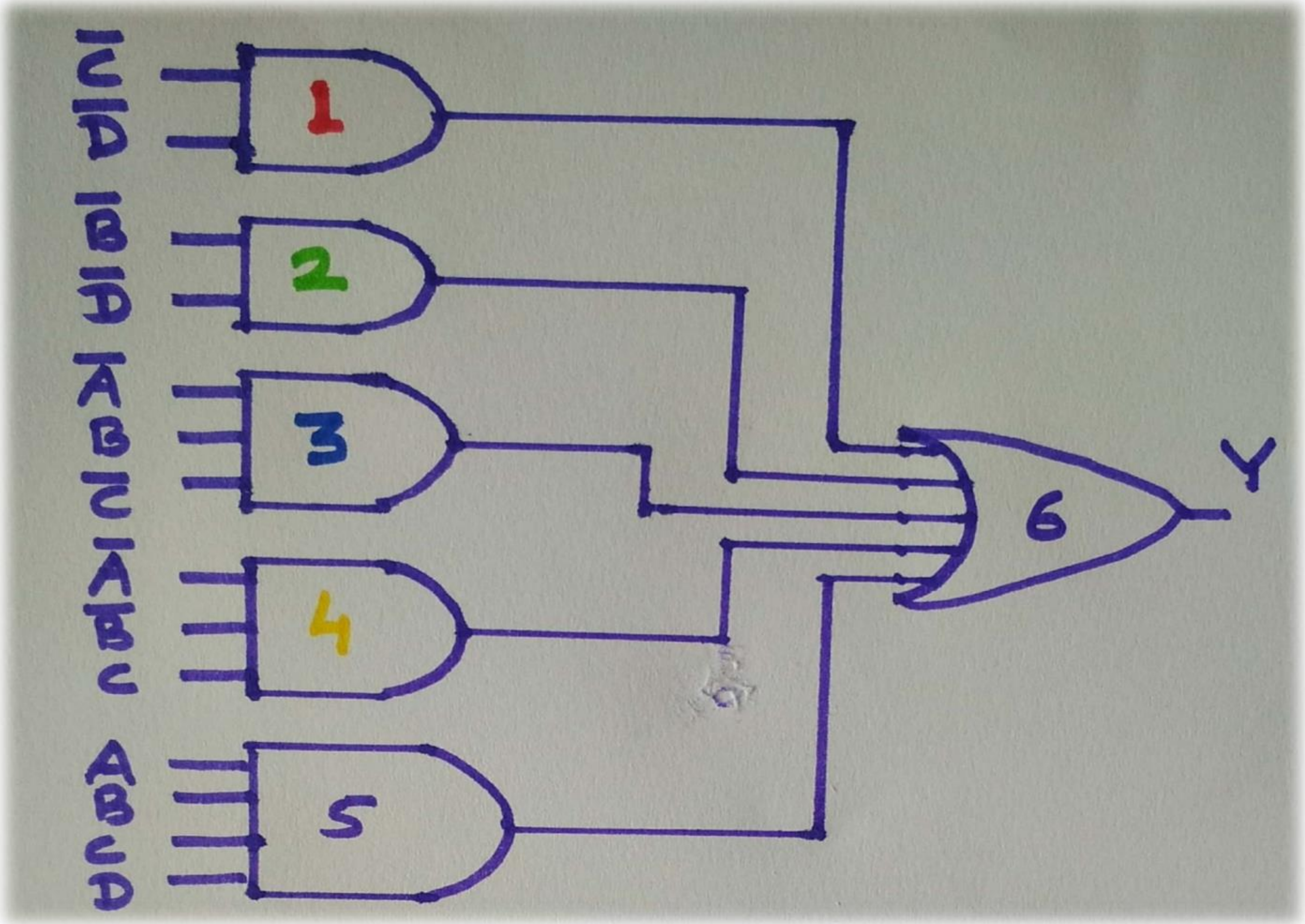
$A\,B\,C\,D$

$\bar{B}\,\bar{D}$

$\bar{A}\,\bar{B}\,C$

$$\therefore\ Y = \bar{C}\,\bar{D} + \bar{B}\,\bar{D} + \bar{A}\,B\,\bar{C} + \bar{A}\,\bar{B}\,C + A\,B\,C\,D$$

$$\therefore\ Y = \overline{C}\,\overline{D} + \overline{B}\,\overline{D} + \overline{A}\,B\,\overline{C} + \overline{A}\,\overline{B}\,C + A\,B\,C\,D$$



# Draw SOP Logic circuit for the following function:

$$Y = f(A, B, C, D) = \Sigma m\,(3, 4, 5, 7, 9, 13, 14, 15) + \Sigma d\,(10)$$

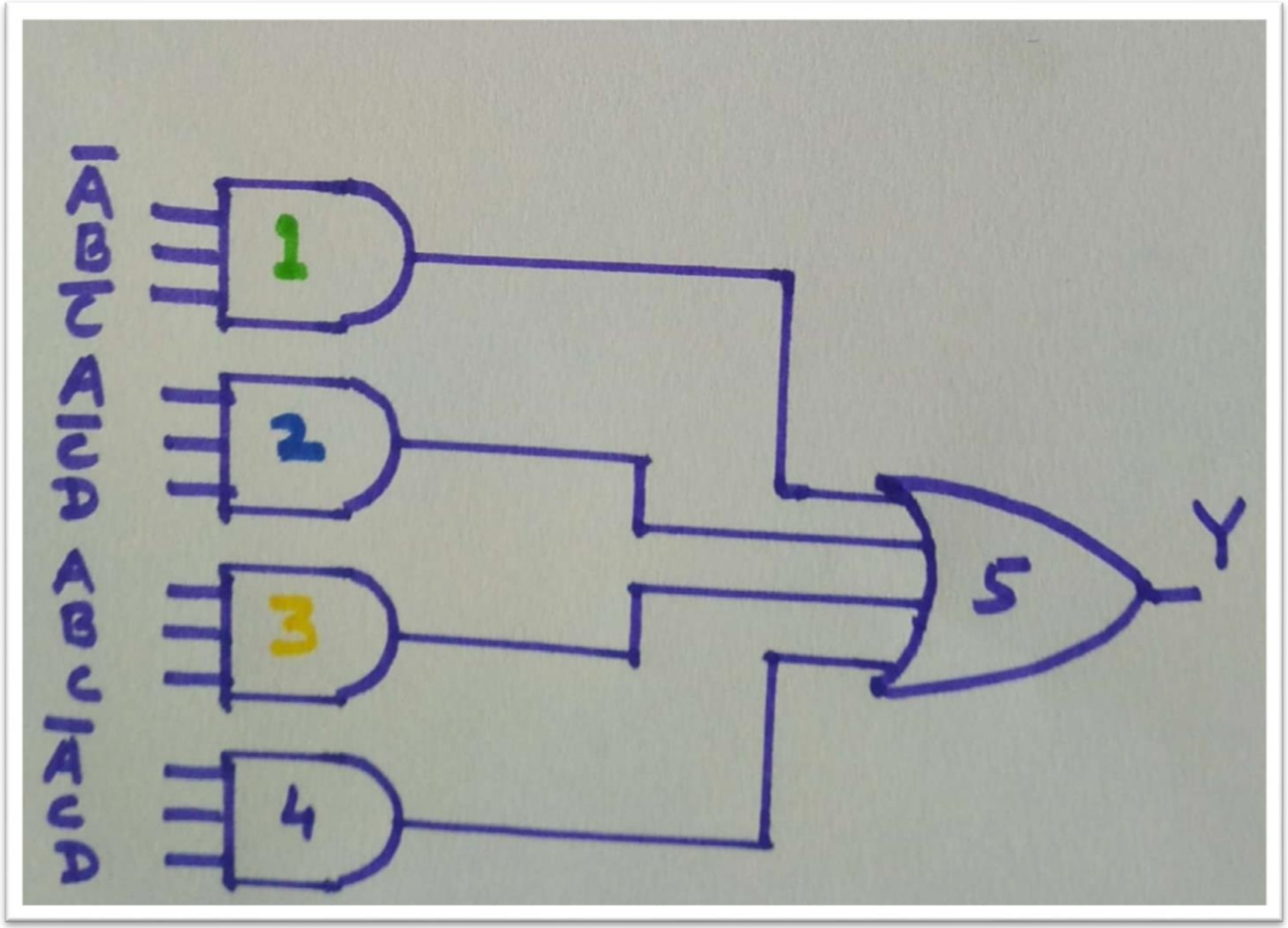
| | $\bar{A}\,\bar{B}$ | $\bar{A}\,B$ | $A\,B$ | $A\,\bar{B}$ |
|---|---|---|---|---|
| $\bar{C}\,\bar{D}$ | **0** | **1** | **0** | **0** |
| $\bar{C}\,D$ | **0** | **1** | **1** | **1** |
| $C\,D$ | **1** | **1** | **1** | **0** |
| $C\,\bar{D}$ | **0** | **0** | **1** | **d** |

**Redundant Group**

**Redundant Group**

$$\therefore\ Y = \bar{A}\,B\,\bar{C} + A\,\bar{C}\,D + A\,B\,C + \bar{A}\,C\,D$$

$$\therefore\ Y = \bar{A}B\bar{C} + A\bar{C}D + ABC + \bar{A}CD$$



# Product of Sum (POS) method

## O/P Y is function of four variables A, B, C and D. (A MSB, D LSB)

| | 0 0 | 0 1 | 1 1 | 1 0 |
|---|---|---|---|---|
| 0 0 | | | | |
| 0 1 | | | | |
| 1 1 | | | | |
| 1 0 | | | | |

| | $A + B$ | $A + \overline{B}$ | $\overline{A} + \overline{B}$ | $\overline{A} + B$ |
|---|---|---|---|---|
| $C + D$ | | | | |
| $C + \overline{D}$ | | | | |
| $\overline{C} + \overline{D}$ | | | | |
| $\overline{C} + D$ | | | | |

# Draw POS Logic circuit for the following function:

## $Y = f(A, B, C, D) = \pi M\,(3, 4, 5, 9, 13, 14) + \pi\, d\,(7, 15)$

| | $A+B$ | $A+\overline{B}$ | $\overline{A}+\overline{B}$ | $\overline{A}+B$ |
|---|---|---|---|---|
| $C+D$ | **1** | **0** | **1** | **1** |
| $C+\overline{D}$ | **1** | **0** | **0** | **0** |
| $\overline{C}+\overline{D}$ | **0** | **d** | **d** | **1** |
| $\overline{C}+D$ | **1** | **1** | **0** | **1** |

Take groups of zeroes. Consider don't care as zeroes to have a bigger group.

**Redundant Group**

$\therefore\ Y = (A+\overline{B}+C)\,(\overline{A}+C+\overline{D})\,(\overline{A}+\overline{B}+\overline{C})\,(A+\overline{C}+\overline{D})$

$$\therefore\ Y = (A+\overline{B}+C)\,(\overline{A}+C+\overline{D})\,(\overline{A}+\overline{B}+\overline{C})\,(A+\overline{C}+\overline{D})$$